



Monitoring of bioremediation by soil biological activities

R. Margesin*, A. Zimmerbauer, F. Schinner

Institute of Microbiology (N.F.), University of Innsbruck, Technikerstrasse 25, A-6020 Innsbruck, Austria

Received 3 March 1999; accepted 14 April 1999

Abstract

An evaluation of soil biological activities as a monitoring instrument for the decontamination process of a mineral-oil-contaminated soil was made using measurements of microbial counts, soil respiration, soil biomass and several enzyme activities. The correlations between these parameters and with the levels of hydrocarbon residues were investigated; the effects of different N- and P-sources on hydrocarbon decontamination and soil biological activities were determined. Inorganic nutrients stimulated hydrocarbon biodegradation but not all biological activities to a significant extent. Biodegradation could be monitored well by soil biological parameters: the residual hydrocarbon content correlated positively with soil respiration, biomass-C (substrate-induced respiration), and with activities of soil dehydrogenase, urease and catalase. Soil lipase activity and the number of hydrocarbon utilizers correlated negatively ($P < 0.0001$) with the remaining hydrocarbon content. © 1999 Elsevier Science Ltd. All rights reserved.

Keywords: Mineral oil (petroleum) hydrocarbons; Biodegradation; Bioremediation; Monitoring; Soil respiration; Biomass; Dehydrogenase; Catalase; Urease; Lipase

1. Introduction

Mineral oil hydrocarbons are the most frequently occurring environmental contaminants. The biological decontamination became a valuable alternative to chemical and biological treatments with the increasing attention towards the preservation of the environment (Norris, 1994; Allard and Neilson, 1997; Margesin and Schinner, 1999). Hydrocarbon-degrading microorganisms are ubiquitous in most ecosystems where contaminants may serve as organic carbon sources. In most environments, enrichments of oil-degrading microbial communities occur soon after contamination (Margesin and Schinner, 1999; Atlas and Bartha, 1992). The intensity of hydrocarbon biodegradation in soil is influenced by several environmental factors such as involved

indigenous microbial populations, nutrient availability, oxygen, pH, temperature, water content, quality, quantity and bioavailability of contaminants, and soil properties. In many contaminated areas, the conditions for microbial degradation may be distinctly limited. Bioremediation techniques accelerate the naturally occurring biodegradation by optimizing conditions for biodegradation through aeration, addition of nutrients and control of pH and temperature (Norris, 1994; Atlas and Bartha, 1992; Morgan and Watkinson, 1989).

To assess the results of biological decontamination it is not only sufficient to measure the remaining hydrocarbon content in soil, it is necessary to observe also the microbial processes (Joergensen et al., 1995) because not all mineral oil hydrocarbons are completely mineralized to CO₂ in the course of biodegradation.

Soil biological investigations (such as soil respiration, biomass, enzyme activities, microbial counts) can give information on the presence of viable microorganisms as well as on the intensity and on the kind and duration of the effects of pollutants on the metabolic activity of soil,

* Corresponding author. Tel.: +43-512/507-6021; fax: +43/512/507-2929.

E-mail address: rosa.margesin@uibk.ac.at (R. Margesin).

and are thus well suited to measure the impact of pollution on soil health (Schinner et al., 1996; van Beelen and Doelman, 1997). Soil biological methods used to describe microbial activities in the course of bioremediation were mainly restricted to soil respiration, dehydrogenase activity and microbial counts (Song and Bartha, 1990; Janke et al., 1992; van der Waarde et al., 1995; Margesin and Schinner, 1997).

The objective of this study was the evaluation of several soil biological methods with regard to their usefulness for monitoring and for assessing the results of the bioremediation process in a soil experimentally contaminated with diesel oil. Changes in hydrocarbon concentration and the accompanying changes in microbial activities with time were measured. The effect of different nitrogen and phosphorus sources on hydrocarbon decontamination and soil biological activities was determined.

2. Materials and methods

2.1. Soil

The soil investigated was an uncontaminated arable soil sampled from the A-horizon in Vill/Innsbruck (Tyrol, Austria) at 840 m a.s.l. The bedrock of the soil was alluvial moraine gravel (calcium carbonate/silicate); the soil type was pararendzina. The soil had a pH of 6.9 (measured in CaCl_2), and contained 0.4% CaCO_3 , 2.6% humus, 0.14% total nitrogen (Kjeldahl) and 0.001% P_2O_5 (CAL/DL). The particle size distribution was 50.1% sand ($> 60 \mu\text{m}$), 19.7% coarse silt ($20\text{--}60 \mu\text{m}$), 20.5% fine and medium silt ($2\text{--}20 \mu\text{m}$), and 9.7% clay ($< 2 \mu\text{m}$).

2.2. Bioremediation experiment

Seven pans were prepared, each containing 8 kg of sieved ($< 4 \text{ mm}$) soil, 6 g $\text{CaCO}_3 \text{ kg}^{-1}$ soil in order to maintain the soil pH in the neutral to slightly alkaline pH during the experiment (Scheffer and Schachtschabel, 1979) and 1000 ml of distilled water. After 3 days at 20°C , each pan was contaminated with 5000 mg of diesel oil (85.7% C, density 820 g l^{-1}) kg^{-1} soil dry weight (dw) and treated as follows:

- Pan 1 (poisoned control) contained 5 g sodium azide kg^{-1} soil dw to inhibit soil microorganisms and to monitor abiotic hydrocarbon elimination. In order to maintain the inhibiting effect, sodium azide was re-added after 13, 36, 56 and 82 days of incubation.
- Pan 2 contained no further supplementation in order to evaluate hydrocarbon degradation by the indigenous soil microorganisms.
- Pans 3–7 were supplemented with different N- and P-sources:

- Pan 3 contained an agricultural N–P–K fertilizer [15% N (9.5% $\text{NH}_3\text{-N}$, 5.5% $\text{NO}_3\text{-N}$), 5.5% P, 12.4% K] suspended in sterile distilled water; the C:N ratio was adjusted to 20:1 whereby the C-concentration was related to the hydrocarbon contamination.
- Pan 4 contained the fertilizer calcium ammonium nitrate [27% N (5.5% $\text{NH}_3\text{-N}$, 13.5% $\text{NO}_3\text{-N}$)], suspended in sterile water, at a C:N ratio of 20:1.
- Pan 5 contained the same N-source as pan 4, additionally the N:P ratio was adjusted to 10:1 by adding K_2HPO_4 dissolved in sterile water.
- Pan 6 contained urea (47% N), dissolved in sterile water, at a C:N ratio of 20:1.
- Pan 7 contained the same urea concentration as pan 6, and K_2HPO_4 at a N:P ratio of 10:1.

The water content was adjusted in all seven pans to $\sim 50\%$ of the maximum soil water holding capacity. The pans were covered with perforated paper lids and incubated in the dark at 20°C and 70% relative humidity for 88 days. Sufficient oxygen was provided by thorough mixing of the pan contents every second day. The soil water content was checked and adjusted regularly.

In regular time intervals, i.e., after 4, 11, 18, 26, 32, 38, 53, 67 and 88 days, chemical, physical and biological analyses were performed with soil samples from each pan.

2.3. Chemical and physical analyses

Soil dry weight and soil pH were determined as described (Schinner et al., 1996).

The residual soil hydrocarbon content was measured with four replicates according to the German standard method DIN 38 409-H18 (1980). After extraction of 25 g soil with 1,1,2-trichloro-trifluoro-ethane, the hydrocarbon content was quantified by infrared spectroscopy (Margesin and Schinner, 1997).

2.4. Soil biological analyses

The following soil biological analyses were performed with four replicates as described (Schinner et al., 1996): biomass-C by substrate-induced respiration, ninhydrin-reactive biomass-N by fumigation–extraction, soil respiration (CO_2 -evolution) by Isermeyer technique, dehydrogenase activity with the substrate INT, catalase activity, lipase activity by titration, urease activity by colorimetry. From pre-experiments it was clear that the presence of diesel oil did not influence the analytical methods applied in this study.

Soil microbial counts were determined by the plate-count method for viable cells. Soil suspensions were prepared by shaking soil corresponding to 10 g soil dw with 100 ml of sterile sodium pyrophosphate (0.2%) for 30 min at 20°C and 150 rpm. 0.1 ml of appropriate

dilutions of the soil suspensions, prepared in 0.9%NaCl, was surface spread onto agar plates. R₂A-agar plates (Reasoner and Geldreich, 1985). were used to determine the total heterotrophic microorganisms. Hydrocarbon-degrading microorganisms were quantified by using oil-agar plates containing diesel oil as the sole carbon source (Günther et al., 1994; Margesin and Schinner, 1998). All enumerations were performed with three replicates. Colony-forming units (cfu) were counted after seven days at 20°C.

2.5. Statistical data analyses

Normal distribution of the data was confirmed by the Kolmogorov-Smirnow-test. Data were analysed by analysis of variance ($P < 0.05$), by multiple range analyses [least significant difference (LSD), $P < 0.05$], and by correlation analyses. Treatments were considered significant when $P < 0.05$.

3. Results

3.1. Hydrocarbon decontamination

After 88 days at 20°C, the initial hydrocarbon contamination of 5000 mg kg⁻¹ soil dw was reduced to 609–1328 mg kg⁻¹ dw in pans 2–7, corresponding to decontamination percentages of 88–73%; whereby 35% of the hydrocarbon elimination (3232 mg kg⁻¹ dw) had to be attributed to abiotic processes (poisoned control, pan 1) such as volatilization, sorption processes and chemical transformations (Fig. 1).

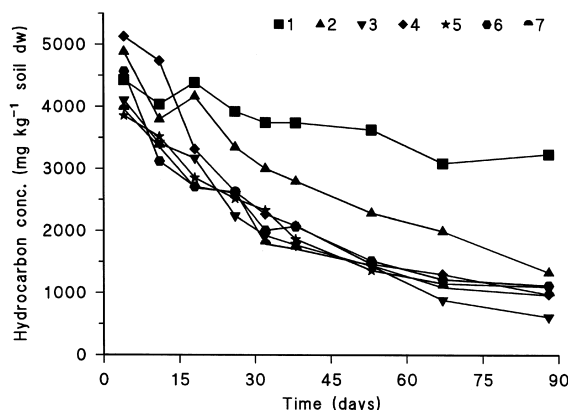


Fig. 1. Bioremediation of a soil contaminated with diesel oil (5000 mg kg⁻¹ soil dw) during 88 days. Values are the means of four replicates (SD $\pm$ 15% or less). (1) poisoned control; (2) unfertilized soil; (3) soil + N-P-K fertilizer; (4) soil + calcium ammonium nitrate; (5) soil + calcium ammonium nitrate + K₂HPO₄; (6) soil + urea; (7) soil + urea + K₂HPO₄.

Both treatment and time had a significant influence on soil decontamination. Hydrocarbon degradation was significantly lower in unfertilized (pan 2) than in fertilized soils (pans 3–7). Beside of nitrogen, also phosphorus enhanced bioremediation significantly and independent of the nitrogen source. There was no significant difference between decontamination in soils containing both a N- and a P-source (pans 3, 5 and 7) (Table 1). Decontamination results were confirmed by gaschromatographic analyses (data not shown).

3.2. Soil pH

The soil pH remained in the neutral to slightly alkaline range during the whole incubation period, which is favourable for bioremediation (Atlas and Bartha, 1992; Morgan and Watkinson, 1989; Margesin and Schinner, 1997). The pH was significantly higher in urea-containing soils (pans 6 and 7) than in soils containing other N-sources, such as NO₃-N and NH₃-N (pans 3 and 4) (Table 1).

3.3. Soil microbial counts

Microbial counts were significantly lower in the poisoned control than in pans 2–7. The number of heterotrophic microorganisms remained almost constant during the whole incubation period, whereas the number of hydrocarbon-degrading microorganisms increased with time. Already after 21 days, the counts of diesel-oil-utilizers corresponded to the one of heterotrophs, thus showing that soil microorganisms adapted rapidly to the hydrocarbon contamination and were able to utilize diesel oil as the sole carbon source. However, the absence or presence of nutrients did not influence significantly the counts of hydrocarbon degraders (Table 1).

3.4. Soil respiration (CO₂-evolution) and soil biomass

CO₂-evolution in the poisoned control was attributed to release of CO₂ from carbonate, but the abiotic CO₂-evolution is also known from non-calcareous soils where abiotically formed CO₂ is produced by chemical decarboxylation of organic matter (Schinner et al., 1996).

Soil respiration (in terms of CO₂-evolution) in pans 2–7 was significantly higher than the one in the poisoned control and reached maximum values after 11 days, decreasing thereafter and remaining almost constant. Biomass-C (determined by glucose-induced respiration) showed maximum values after 25 days, but, contrary to soil respiration, was characterized by some fluctuations, i.e., a repeatedly decrease and increase due to microbial succession. Both respiration and biomass-C were significantly higher in the urea-containing soils (pans 6 and 7) than in other soil treatments (Table 1).

Table 1

Effect of fertilization on hydrocarbon decontamination, soil pH and soil biological parameters during 88 days (values are means of nine data points from a time series (each data point represents the mean of four replicates))

Parameter	Soil treatment ^a						
	1	2	3	4	5	6	7
Hydrocarbons (mg kg ⁻¹ dw)	3800 d ^b	3064 c	2171 a	2654 b	2280 a	2322 ab	2175 a
Soil pH	7.73 c	7.51 ab	7.45 a	7.47 a	7.52 ab	7.60 b	7.62 b
Hydrocarbon utilizers [log (cfu g ⁻¹ dw)]	2.12 a	6.66 b	6.79 b	6.77 b	6.56 b	6.60 b	6.93 b
Respiration (μg CO ₂ g ⁻¹ dw 24 h ⁻¹)	114 a	170 b	211 bcd	189 bc	226 cd	250 de	291 e
Biomass-C (μg C _{mic} g ⁻¹ dw)	188 a	313 b	301 ab	312 b	376 bc	466 cd	538 d
qCO ₂ (μg CO ₂ -C mg ⁻¹ C _{mic})	20.21 a	15.76 a	20.82 a	112.38 b	48.77 ab	25.34 a	25.72 a
Biomass-N (μg g ⁻¹ dw)	0.79 a	35.72 c	23.58 b	24.92 b	21.85 b	24.24 b	18.26 b
Dehydrogenase activity (μg INTF g ⁻¹ dw)	6.36 a	158.83 cd	169.54 d	134.31 bc	146.91 bcd	130.82 b	145.36 bcd
Catalase activity (catalase units)	0.08 a	1.26 b	2.25 c	2.20 c	2.25 c	2.93 c	2.88 c
Lipase activity (lipase units)	0.52 a	2.09 b	3.97 d	3.04 c	3.22 c	3.90 d	4.06 d
Urease activity (μg NH ₄ -N g ⁻¹ dw)	8.42 a	50.61 b	47.16 b	54.52 b	54.13 b	46.72 b	51.89 b

^a (1) Poisoned control; (2) soil without nutrients; (3) soil + N–P–K fertilizer; (4) soil + calcium ammonium nitrate; (5) soil + calcium ammonium nitrate + K₂PO₄; (6) soil+urea; (7) soil + urea + K₂HPO₄.

^b Different letters in a row indicate statistically significant differences (LSD, $P < 0.05$) between soil treatments; the same letters indicate that the values are not significantly different.

The metabolic quotient $q\text{CO}_2$ was calculated from respiration and biomass-C (Schinner et al., 1996). The lower the $q\text{CO}_2$, the more efficient is the microbial turnover. The $q\text{CO}_2$ remained unaffected by the different soil treatments, except in soils containing calcium ammonium nitrate as nitrogen source which showed a significantly higher $q\text{CO}_2$ (Table 1).

Biomass-N showed a similar time course like biomass-C in all fertilized soils, i.e., a decrease after 25 days. Remarkably, biomass-N in the unfertilized soil (pan 2) increased continuously until day 66, and was significantly higher in the unfertilized than in fertilized soils, independent of the nutrients (Table 1).

3.5. Soil enzymes

Like all soil biological parameters, also soil enzyme activities measured in this study were significantly lower in the poisoned control than in all other soil treatments (Table 1). Activities of soil dehydrogenase, catalase and urease increased immediately after oil contamination, but declined with decreasing hydrocarbon content. Comparable to soil biomass and respiration, also soil dehydrogenase, catalase and urease showed fluctuations, especially during the first weeks; maximum values were observed after 11–25 days.

Catalase activity was significantly lower in the unfertilized than in fertilized soils, but the composition of nutrients had no significant influence on this enzyme activity. On the contrary, urease activity was not significantly influenced by the presence or absence of nutrients; a similar trend was observed for dehydrogenase activity (Table 1).

Lipase activity showed an inverse time course like hydrocarbon decontamination, i.e., the enzyme activity

increased with decreasing hydrocarbon content. After an initial rapid increase in lipase activity in all fertilized soils (pans 3–7), the enzyme activity remained level or decreased slightly. Lipase activity increased much slower in the unfertilized than in fertilized soils (Fig. 2). Nutrients increased lipase activity significantly; calcium ammonium nitrate, independent of the presence or absence of phosphorus, stimulated lipase activity to a significantly lower extent than did the other nitrogen sources tested (Table 1).

3.6. Effect of fertilization

Analyses of variance and multiple range analyses (Table 1) showed that all biological analyses were

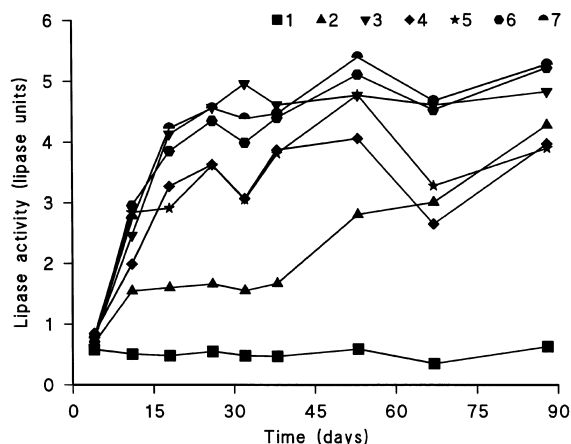


Fig. 2. Soil lipase activity during bioremediation of a diesel-oil-contaminated soil. Values are the means of four replicates (SD $\pm$ 15% or less). For symbols, see Fig. 1.

significantly lowest in the poisoned control, a proof for the inhibiting effect of NaN_3 . The presence or absence of nutrients had no significant effect on urease activity and on the number of hydrocarbon utilizers. The absence of nutrients resulted in a significantly higher biomass-N than did fertilization. Catalase activity was significantly higher in fertilized than in unfertilized soils, but the nutrient source was without any effect. The metabolic quotient $q\text{CO}_2$ was significantly highest in soils containing calcium ammonium nitrate. The presence of urea resulted in a significant increase in soil respiration, biomass-C and lipase activity, but did not influence urease activity.

3.7. Correlations

The poisoned control was not included in correlation analyses, only pans 2–7 were considered. These analyses demonstrated that several of the investigated soil biological parameters correlated significantly with the residual soil hydrocarbon content in unfertilized and fertilized soils during the investigated period of 88 days (Table 2). Hydrocarbon content correlated strongly negatively with the number of hydrocarbon utilizers, with lipase activity and soil pH ($P < 0.0001$), and correlated positively with soil respiration and biomass-C ($P < 0.001$), and with activities of soil dehydrogenase, urease ($P < 0.01$) and catalase ($P < 0.05$). No correlation was observed between hydrocarbon content and metabolic quotient or biomass-N.

There were also several significant correlations between the other measured soil parameters (Table 2). The soil pH correlated significantly with soil biomass-C and biomass-N as well as with activities of dehydrogenase, urease and lipase. The number of hydrocarbon utilizers correlated significantly negatively with soil respiration, biomass-C and catalase activity, and significantly positively with lipase activity. Soil respiration (CO_2 -evolution) correlated significantly with biomass-C (the substrate-induced respiration), but also with all measured enzyme activities, whereas biomass-C correlated only with lipase and catalase activity among the measured enzymes. The metabolic quotient correlated only with biomass-C and with the soil pH, but with none of the other parameters tested. Activities of dehydrogenase, catalase, urease and soil respiration correlated significantly with each other, but not with lipase activity.

4. Discussion

The present study reports the reaction of various soil biological activities during the microbial decontamination of a soil experimentally contaminated with diesel oil, with and without biostimulation of the indigenous microorganisms by bioremediation treatments using

different N- and P-sources. We demonstrated that under favourable conditions, biodegradation can restore oil-polluted soil, and inorganic nutrient amendment can accomplish this. The input of large quantities of organic carbon sources tends to result in a rapid depletion of available inorganic nutrients such as nitrogen and phosphorus. Consequently, rates of biodegradation are limited. Inorganic nutrient-amendment increased oil-biodegradation rates in both marine and terrestrial ecosystems considerably (Margesin and Schinner, 1999; Atlas and Bartha, 1992; Morgan and Watkinson, 1989). We have shown in this study that it is important to add both nitrogen and phosphorus. The addition of phosphorus can stimulate the biodegradation of oil hydrocarbons, but various P-sources can have different effects on biodegradation, depending on the solubility of the P-sources and their toxicity (Mills and Frankenberger, 1994). Nevertheless, the soil hydrocarbon content cannot be reduced to zero by using biological decontamination, even after a prolonged treatment (Margesin and Schinner, 1997). The remaining 10–30% consist of hydrocarbons that are structurally less available for biodegradation due to their recalcitrance and very limited bioavailability (Allard and Neilson, 1997; Margesin and Schinner, 1999), causing a downward trend in microbial abundance well in advance of the exhaustion of all hydrocarbons (Song and Bartha, 1990).

The soil microorganisms seemed to tolerate and utilize the new carbon source quickly, as demonstrated by increasing microbial counts, soil respiration, biomass and enzyme activities, especially during the first weeks of the study. With a decreased residual hydrocarbon content, also the measured biological parameters – except microbial counts and lipase activity – were reduced in both unfertilized and fertilized soils. This can be attributed to a reduced bioavailability of carbon sources, and most probably also to the accumulation of recalcitrant and/or toxic compounds in the course of biodegradation.

It is well-known that oil contamination causes significant changes in microbial populations. Many authors observed an increase in the number of hydrocarbon utilizers with time during biological soil decontamination, correlating with biodegradation and biological activities (Song and Bartha, 1990; Sextone et al., 1978). We observed in this study that the number of hydrocarbon degraders increased after stimulation of the biodegradation process, but this increase was also present in soil without nutrient amendments. The same result was obtained by van der Waarde et al. (1995) who assumed that changes in bacterial numbers are indicative of a stimulated biodegradation process, but do not represent an accurate measurement of the actual biodegradation. The quantification of viable soil microorganisms alone gives no information about the efficiency of the populations, and it is well known that only a small

Table 2
Correlation matrix (coefficients and significance levels) between hydrocarbon decontamination, soil pH and soil biological parameters during 88 days^a

	Hydroc. concentr.	pH	Hydroc. utilizers	Respiration	Biomass-C	<i>q</i> CO ₂	Biomass-N	Dehydro- genase	Catalase	Lipase
Soil pH	-0.5543 0.0000****	/								
Hydrocarbon utilizers	-0.8273 0.0000****	0.2090 0.2212	/							
Respiration (CO ₂ -evolution)	0.5086 0.0001***	-0.2567 0.0610	-0.6744 0.0000****	/						
Biomass-C	0.4938 0.0001***	-0.5048 0.0001****	-0.5876 0.0002***	0.5553 0.0000****	/					
<i>q</i> CO ₂	-0.1793 0.1945	0.2978 0.0288*	-0.0342 0.8430	0.0447 0.4780	-0.3074 0.0238*	/				
Biomass-N	0.2116 0.1488	-0.0893 0.5461	-0.2198 0.2432	-0.1693 0.2501	0.2375 0.1041	-0.0640 0.6394	/			
Dehydrogenase activity	0.3821 0.0044**	-0.5060 0.0001****	-0.0265 0.8779	0.3060 0.0244*	0.0939 0.4994	-0.1846 0.1816	0.1534 0.2980	/		
Catalase activity	0.3138 0.0208*	-0.2568 0.0609	-0.3662 0.0280*	0.6769 0.0000****	0.4680 0.0004****	-0.0498 0.7206	-0.0024 0.9870	0.4942 0.0001***	/	
Lipase activity	-0.8209 0.0000****	0.5025 0.0001****	0.6908 0.0000****	-0.2819 0.0389*	-0.2777 0.0420*	0.1181 0.3949	-0.1798 0.2214	-0.2186 0.1123	0.0123 0.9298	/
Urease activity	0.3893 0.0036**	-0.4939 0.0001****	0.0048 0.9780	0.4440 0.0008****	0.2134 0.1213	0.0333 0.8110	-0.0893 0.5461	0.5340 0.0000****	0.4291 0.0012	-0.2332 0.0897

^aThe poisoned control was excluded from calculations; sample number = 54 (values were determined with nine data points and six soil treatments; each data point represents the mean of four replicates; * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$).

part of soil microorganisms can be isolated and cultivated on laboratory media. Therefore, the quantification of soil biological activities is often used to interpret the intensity of microbial metabolism in soil (Schinner et al., 1996; van Beelen and Doelman, 1997; Kreysa and Wiesner, 1995).

Respirometric measurements were used repeatedly as a relevant parameter during microbial oil decontamination (Janke et al. 1992; van der Waarde et al., 1995; Margesin and Schinner, 1997; Kreysa and Wiesner, 1995). An increased soil dehydrogenase (Janke et al., 1992; Margesin and Schinner, 1997; Frankenberger and Johanson, 1982; Kandeler et al., 1986) and catalase activity (Janke et al., 1992) – both enzymes are involved in intracellular metabolism – in oil-polluted soils was attributed to the adaptation and reproduction of the hydrocarbon degraders. In an earlier study (van der Waarde et al., 1995) the parameter that had the best correlation with hydrocarbon removal and soil respiration was dehydrogenase activity.

To our knowledge, no data on activities of lipase (involved in lipid degradation) and urease (involved in the N-cycle) in oil-contaminated soils are available from the literature. We could demonstrate for the first time that soil lipase activity is an excellent indicator for monitoring oil decontamination. We assume that enzyme systems for lipid degradation are similar to those involved in biodegradation of mineral oil hydrocarbons. Contrary to activities of soil dehydrogenase, catalase and urease, which increased immediately after oil contamination and decreased at low hydrocarbon concentrations, lipase activity was low at high hydrocarbon contents but stimulated with decreasing contaminant content.

Soil microbial biomass is an important indicator of the status of soil vitality and is often determined by the short-term substrate (glucose)-induced respiration. This method is recommended to determine the effects of organic compounds on the soil microflora (Schinner et al., 1996; Kreysa and Wiesner, 1995) and was used for oil-contaminated soils (Kandeler et al., 1994). However, this method registers only glucose-utilizing microorganisms. Big part of the hydrocarbon utilizers was reported to be able to utilize glucose (Song et al., 1986). Because of the above-mentioned problem concerning biomass estimation, the $q\text{CO}_2$, i.e., the respiratory CO_2 -release per unit biomass, has to be interpreted critically. The $q\text{CO}_2$ was evaluated as a moderately sensitive indicator of the effects of pollutants (van Beelen and Doelman, 1997).

In conclusion, soil biological methods should complement chemical methods when evaluating and assessing the success of decontamination. As earlier suggested (Song and Bartha, 1990) the correlations between the soil biological parameters and with the levels of hydrocarbon residues could help in elucidating microbial contributions to oil elimination from soil.

Acknowledgements

This study was supported by the Land Tyrol and by the Austrian Federal Ministries of Science and Traffic, and of Environment, Youth and Family.

References

- Allard, A.S., Neilson, A.H., 1997. Bioremediation of organic waste sites: a critical review of microbiological aspects. *Int. Biodeterioration Biodegradation* 39, 253–285.
- Atlas, R.M., Bartha, R., 1992. Hydrocarbon biodegradation and oil spill bioremediation. *Adv. Microbial Ecol.* 12, 287–338.
- DIN 38 409-H18, 1980. Deutsche Einheitsverfahren zur Wasser-, Abwasser-und Schlammuntersuchung. Summarische Wirkungs-und Stoffkenngrößen (Gruppe H). Bestimmung von Kohlenwasserstoffen (H18). Deutsches Institut für Normung e.V., Berlin.
- Frankenberger, W.T., Johanson, J.B., 1982. Influence of crude oil and refined petroleum products on soil dehydrogenase activity. *J. Environ. Qual.* 11, 602–607.
- Günther, T., Sack, U., Schlosser, D., Fritsche, W., 1994. Anreicherung und Isolierung von Umweltchemikalien abbauenden Mikroorganismen. In Alef, K. (Ed.), *Biologische Bodensanierung*. Verlag Chemie, Weinheim, pp. 77–102.
- Janke, S., Schamber, H., Kunze, C., 1992. Beeinflussung der bodenbiologischen Aktivität durch Heizöl. *Angew. Botanik* 66, 42–45.
- Joergensen, R.G., Schmaedeke, F., Windhorst, K., Meyer, B., 1995. Biomass and activity of microorganisms in a fuel-oil contaminated soil. *Soil Biol. Biochem.* 27, 1137–1143.
- Kreysa, G., Wiesner, J. (Eds.), 1995. Bioassays for soils (Biologische Testmethoden für Böden). Dechema e.V., Frankfurt a. M.
- Kandeler, E., Pennerstorfer, C., Bauer, E., Braun, R., 1994. Microbiological control of the biological decontamination of soils. *Z. Pflanzenernähr. Bodenkd.* 157, 345–350.
- Margesin, R., Schinner, F., 1997. Bioremediation of diesel-oil-contaminated alpine soils at low temperatures. *Appl. Microbiol. Biotechnol.* 47, 462–468.
- Margesin, R., Schinner, F., 1998. Low-temperature-bioremediation of a waste water contaminated with anionic surfactants and fuel oil. *Appl. Microbiol. Biotechnol.* 49, 482–486.
- Margesin, R., Schinner, F., 1999. Biological decontamination of oil spills in cold environments. *J. Chem. Technol. Biotechnol.* 74, 1–9.
- Mills, S.A., Frankenberger, W.T., 1994. Evaluation of phosphorus sources promoting bioremediation of diesel fuel in soil. *Bull. Environ. Contam. Toxicol.* 53, 280–284.
- Morgan, P., Watkinson, R.J., 1989. Hydrocarbon degradation in soils and methods for soil biotreatment. *CRC Crit. Rev. Biotechnol.* 8, 305–333.
- Norris, R.D. (Ed.), 1994. *Handbook of Bioremediation*. CRC Press, Boca Raton, Florida.
- Reasoner, D.J., Geldreich, E.E., 1985. A new medium for the enumeration and subculture of bacteria from potable water. *Appl. Environ. Microbiol.* 49, 1–7.
- Scheffer, F., Schachtschabel, P., 1979. *Lehrbuch der Bodenkunde* (10. Auflage). Ferdinand Enke, Stuttgart.

- Schinner, F., Öhlinger, R., Kandeler, E., Margesin, R. (Eds.), 1996. *Methods in Soil Biology*. Springer, Heidelberg.
- Sextone, A.J., Everett, K., Jenkins, T., Atlas, R., 1978. Fate of crude and refined oils in north slope soils. *Arctic* 31, 339–347.
- Song, H.G., Bartha, R., 1990. Effects of jet fuel on the microbial community of soil. *Appl. Environ. Microbiol.* 56, 646–651.
- Song, H.G., Pedersen, T.A., Bartha, R., 1986. Hydrocarbon mineralization in soil: relative bacterial and fungal contribution. *Soil Biol. Biochem.* 18, 109–111.
- van Beelen, P.V., Doelman, P., 1997. Significance and application of microbial toxicity tests in assessing ecotoxicological risks of contaminants in soil and sediment. *Chemosphere* 34, 455–499.
- van der Waarde, J.J.V., Dijkhuis, E.J., Henssen, M.J.C., Keuning, S., 1995. Enzyme assays as indicators for biodegradation. In van den Brinck, W.J., Bosman, R., Arendt, F. (Eds.), *Contaminated Soil '95*. Kluwer Academic Publishers, Dordrecht, 1995, pp. 1377–1378.